

AI-ASSISTED CODING

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Assignment-19.1

Batch-14

Lab 19 – Code Translation: Converting Between Programming Languages

Task 1: Python to Java Conversion

PROMPT:

Convert this Python class with attributes and methods into an equivalent Java class with proper constructors and data types.

PYTHON CODE:

```
#generate a python class with attributes and methods
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def greet(self):
        return f"Hello, my name is {self.name} and I am {self.age} years old."

    def have_birthday(self):
        self.age += 1
        return f"Happy birthday! You are now {self.age} years old."

# Example usage
person1 = Person("Alice", 30)
print(person1.greet()) # Output: Hello, my name is Alice and I am 30 years old.
print(person1.have_birthday()) # Output: Happy birthday! You are now 31 years old.
```

JAVA CODE:

```
class Person {  
    private String name;  
    private int age;  
  
    public Person(String name, int age) {  
        this.name = name;  
        this.age = age;  
    }  
  
    public String greet() {  
        return "Hello, my name is " + this.name + " and I am " + this.age + " years old.";  
    }  
  
    public String haveBirthday() {  
        this.age++;  
        return "Happy birthday! You are now " + this.age + " years old.";  
    }  
}
```

OUTPUT:

```
Hello, my name is Alice and I am 30 years old.  
Happy birthday! You are now 31 years old.
```

Task 2 : C++ to Python Function Conversion

PROMPT:

Convert this recursive C++ factorial function into a Python function following Pythonic conventions.

C++ CODE:

```
#include <iostream>
using namespace std;
int factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    return n * factorial(n - 1);
}
int main() {
    int number;
    cout << "Enter a number to calculate its factorial: ";
    cin >> number;
    cout << "Factorial of " << number << " is " << factorial(number) << endl;
    return 0;
}
```

PYTHON CODE :

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)
number = int(input("Enter a number to calculate its factorial: "))
print(f"Factorial of {number} is {factorial(number)}")
```

OUTPUT:

```
Enter a number to calculate its factorial: 5
Factorial of 5 is 120
```

Task 3: SQL to Pandas Query

PROMPT:

Convert this SQL query into an equivalent Pandas DataFrame operation in Python.

SQL QUERY:

```
SELECT * FROM EMPLOYEES;
```

CODE :

```
#CONVERT IT INTO A PANDAS DATAFRAME
import pandas as pd
# Assuming you have a connection to your database and have executed the SQL query
# For example, using SQLAlchemy to connect to a database
from sqlalchemy import create_engine
# Create a connection to the database (replace with your own connection string)
engine = create_engine('your_connection_string_here')
# Execute the SQL query and load the results into a pandas DataFrame
query = "SELECT * FROM EMPLOYEES"
df = pd.read_sql_query(query, engine)
# Now 'df' contains all the records from the EMPLOYEES table
```

OUTPUT:

...	id	name	department
0	1	Alice	HR
1	2	Bob	IT
2	3	Charlie	HR

Task 4: Real-Time Application: Multi-Language Snippet Converter

PROMPT:

Translate this code from one programming language to another and ensure both versions produce the same output.

PYTHON CODE:

```
 import requests
from bs4 import BeautifulSoup
url = "https://sraap.in/"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")
for link in soup.find_all("a"):
    print(link.get("href"))
```

JAVASCRIPT CODE:

```
const links = document.querySelectorAll("a");
links.forEach(link => {
    console.log(link.href);
});
```

OUTPUT:

```
index.php
staff_login.php
student_login.php
```